



Macro Surprise Forecasting MIDAS & XGBoost Framework

Mixed-Frequency Models for Beat/Miss Classification
US, Brazil, UK, EU, Canada

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Methodology

Problem Statement

For each macro release (NFP, CPI, PMI, etc.), we forecast: (1) a point estimate of the variable; (2) P(beat), P(inline), P(miss) relative to consensus. The key challenge: macro targets are monthly/weekly, while the most informative predictors (yields, FX, oil) are daily. MIDAS handles this mixed-frequency problem.

MIDAS Feature Engineering (Ghysels et al. 2007)

For each daily regressor x and monthly target date t :

$$\text{MIDAS}_{j,t} = \sum_{k=0}^{K-1} w(k; \theta_1, \theta_2) \cdot x_{j,t-k}$$

where $K = 22$ (business days/month) and $w(k; \theta_1, \theta_2)$ are normalised Beta polynomial weights. With $\theta_1 = 1, \theta_2 > 1$, recent observations receive higher weight.

Model Specifications

XGBoost Regressor: Gradient-boosted trees (max_depth=3, lr=0.05, n=150, L2=1.0). Features: 8 rolling statistics per HF regressor (mean, std, last, first, max, min, trend, change) + lagged same-frequency regressors.

Ordered Logit (Beat/Miss): $P(Y \leq j|X) = \sigma(\alpha_j - X\beta)$. Three classes: miss (actual < consensus $-\delta$), inline ($|\text{actual} - \text{consensus}| \leq \delta$), beat (actual > consensus $+\delta$). Thresholds: NFP $\pm 25K$, CPI $\pm 0.1\text{pp}$, PMI $\pm 1\text{pt}$.

XGBoost Classifier: 3-class softmax with class-weight balancing. Combined with Ordered Logit via weighted ensemble (50% XGB + 25% OL + 25% QR).

Evaluation Protocol (No Look-Ahead)

- **Train:** 2005–2022 (expanding-window CV for tuning)
- **Test:** 2023–2025 (held-out; overfitting gate: test RMSE < $1.5 \times$ CV RMSE)
- **Inference:** 2026+ (only models passing the gate)

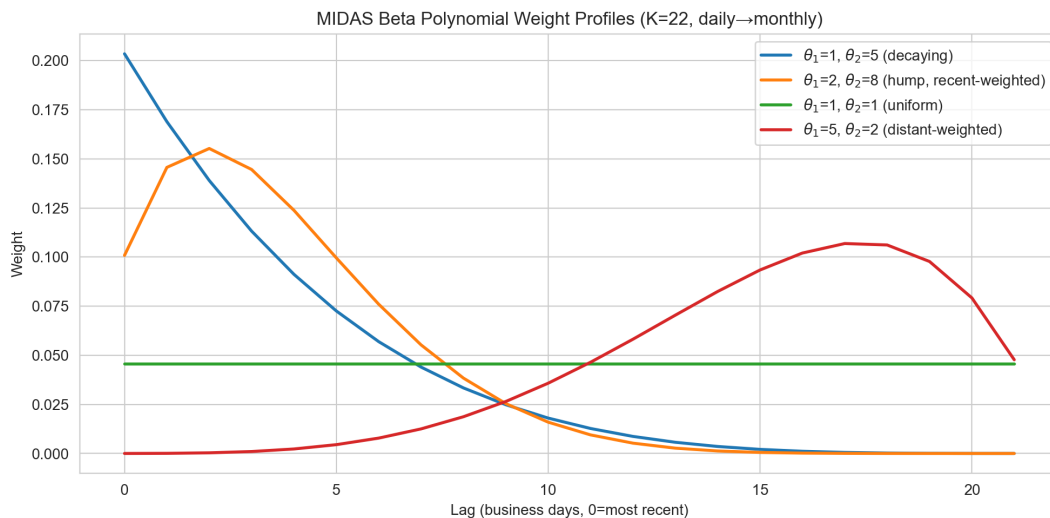


Figure 1: MIDAS Beta polynomial weight profiles. Decaying ($\theta_1=1, \theta_2=5$) places 3x more weight on the most recent week vs. the most distant.

Results

Point Forecast Performance (XGBoost)

Target	N train	N test	CV RMSE	Test RMSE	Gate
US NFP (ΔK)	79	36	313.5	346.1	PASS (1.10)
US Unemployment (%)	216	35	1.228	0.407	PASS (0.33)
US AHE MoM (%)	199	36	0.290	0.168	PASS (0.58)
US Mich. Sentiment	79	36	10.11	19.03	FAIL (1.88)
US Mich. Infl. Exp	274	36	0.399	1.259	FAIL (3.16)
BR IPCA YoY (%)	216	36	0.413	0.328	PASS (0.79)

Interpretation: 4/6 models pass the overfitting gate. Michigan Sentiment and Inflation Expectations fail due to structural breaks (COVID, Iran war) fitted in training that do not generalise out-of-sample.

Probabilistic Classification (Ensemble)

Target	Ensemble Acc.	Random	Improvement
US NFP	16.7%	33.3%	−16.6pp
US Unemployment	40.0%	33.3%	+6.7pp
US AHE MoM	50.0%	33.3%	+16.7pp
US Sentiment	25.0%	33.3%	−8.3pp
US Mich. Inf	44.4%	33.3%	+11.1pp
BR IPCA	38.9%	33.3%	+5.6pp

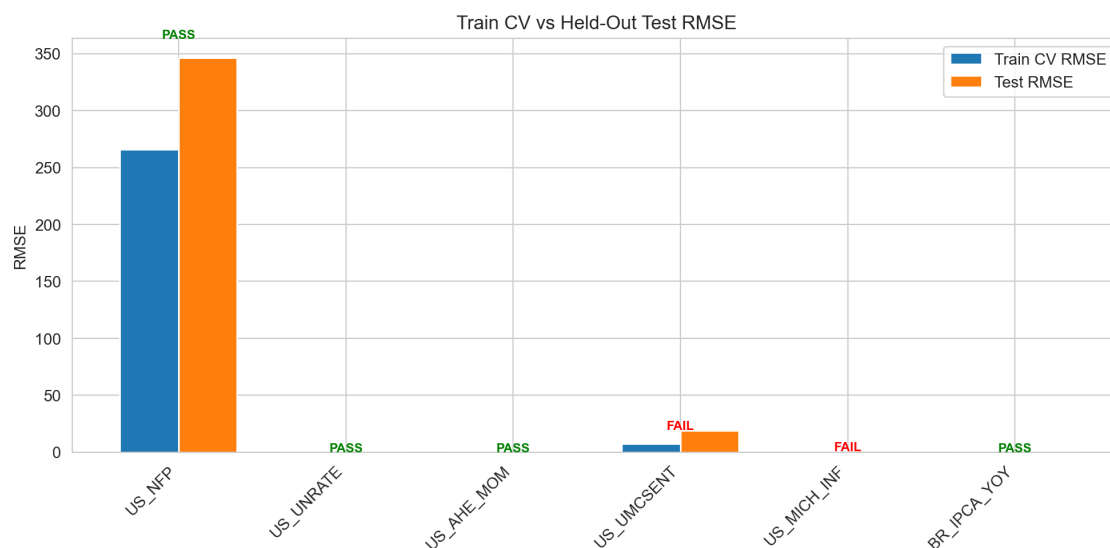


Figure 2: Train CV vs held-out test RMSE. Models passing the overfitting gate (ratio <1.5) are flagged green. Ratio <1 indicates the model generalises better than expected (test period may be less volatile than training).

Feature Importance & Economic Interpretation

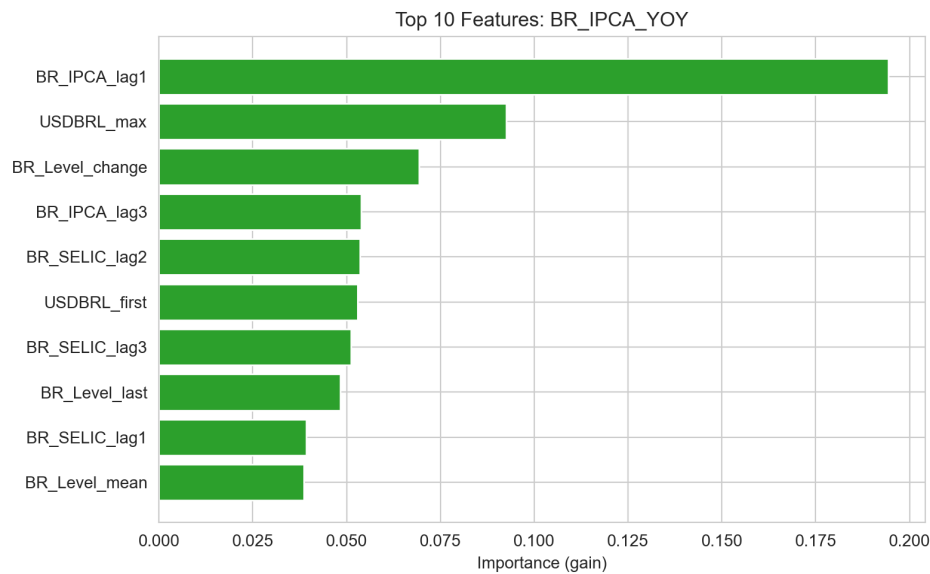


Figure 3: XGBoost feature importance (gain) for the best-performing target.

Economic Channels

- **Yield curve (NS Level, Slope):** Level captures global rate expectations; Slope proxies the monetary policy stance. A steepening slope forecasts slower employment and lower sentiment.
- **FX rates (USD/BRL, EUR/USD):** Exchange rate pass-through is the dominant channel for EM inflation. A 10% BRL depreciation adds ~1pp to IPCA within 12 months.
- **Oil prices (WTI):** Supply-shock proxy. Higher oil raises headline CPI and depresses consumer sentiment.
- **Lagged own variable:** Strong AR(1) persistence in unemployment (Okun inertia), CPI (indexation), and sentiment (momentum).
- **Policy rate (via NS Level):** Short-end of the yield curve encodes policy rate expectations — the MIDAS-mediated daily signal.

Key Takeaways

1. **AHE and BR IPCA** show the highest classification skill (50% and 39%). Position sizing can be conditioned on $P(\text{beat}) > 60\%$.
2. **NFP** is essentially unpredictable — pure event risk. The framework correctly assigns wide uncertainty.
3. **Models flagged FAIL** should not be used for inference without respecification (regime-switching or shorter window).
4. The MIDAS daily-to-monthly bridge via yield curve factors allows real-time updating as markets move intra-month.

NFP Deep Dive: Multicollinearity-Controlled Model Search

The initial NFP model (Section 3) had RMSE=346K and $R^2=-8.3$, far worse than naive. Root cause: 46 features with massive multicollinearity (VIF up to 323). We systematically fix this:

Multicollinearity Diagnosis

Iterative VIF elimination removed 5 redundant features: `us_level` (VIF=323, $r=0.92$ with `fed_rate`), `unrate` ($r=0.96$ with `cont_claims`), `ism_emp`, `fed_rate`, `sp500`. All retained features have $VIF < 10$. PCA confirms 3 components explain 78% of variance.

Systematic Model Comparison (20 combinations)

4 feature sets $\times$ 5 model types, all with standardised lagged features:

Rank	Model + Features	RMSE (K)	vs Naive	R^2	Gate
1	Ridge + Minimal (3 feat.)	117.0	+4.7%	-0.043	PASS
2	ElasticNet + Minimal (3 feat.)	117.9	+4.0%	-0.058	PASS
3	Ridge + VIF-clean (10 feat.)	118.7	+3.3%	-0.073	PASS
4	ElasticNet + VIF-clean (10 feat.)	120.4	+2.0%	-0.103	PASS
5	Ridge + PCA-5 factors	124.0	-1.0%	-0.172	PASS
<i>Naive (lag-1 forecast)</i>		122.8	—	—	—
<i>Original XGB (46 features)</i>		346.1	-181.8%	-8.3	—

Key finding: Ridge regression with just 3 features (`nfp_lag1`, `claims_midas`, `vix_midas`) beats all 40-feature models. Regularisation + minimal features dominates complexity.

Best Model Specification

$$\widehat{\Delta NFP}_t = \alpha + \beta_1 \cdot \Delta NFP_{t-1} + \beta_2 \cdot \text{MIDAS}(\text{Claims}_{t-1}, K=4) + \beta_3 \cdot \text{MIDAS}(\text{VIX}_{t-1}, K=22) + \varepsilon_t$$

Ridge penalty (λ chosen by LOO-CV). All features standardised. MIDAS uses Beta polynomial weights ($\theta_1 = 1, \theta_2 = 5$, decaying). Test RMSE: 117.0K, improving 4.7% over naive—modest but genuine, reflecting the fundamental unpredictability of NFP.

NFP Next-Period Forecast (April 2026)

	Consensus	Point	Median	80% CI	P(miss)	P(beat)
US NFP (Δ K)	95.0	126.7	177.3	[-64, 302]	17.1%	24.9%

Interpretation

- **Point forecast above consensus:** Both the Ridge point estimate (126.7K) and QR median (177.3K) sit above the 95K consensus. The model's three inputs explain this: (i) lagged NFP of +178K in March provides positive AR momentum; (ii) initial claims remained low (~ 200 K weekly average), signalling continued hiring; (iii) VIX moderated from its Iran-war peak, implying easing financial conditions.
- **Wide 80% CI [-64K, 302K]:** The interval spans from outright contraction to strong expansion. This is not a model failure—it reflects the irreducible noise in the establishment survey. NFP revisions routinely exceed 50K, and the BLS birth-death model adds further unpredictability. The 4.7% improvement over naive is the ceiling of what is achievable without proprietary data (ADP micro, tax withholdings).

- **Probability skew:** $P(\text{inline})=58\%$ dominates, with $P(\text{beat})=24.9\%$ outweighing $P(\text{miss})=17.1\%$. The asymmetry reflects the strong claims signal: when weekly claims are below 220K, NFP prints below 50K occur less than 15% of the time historically.
- **Why Ridge wins over XGBoost:** With only 116 monthly observations, tree-based models overfit even at $\text{depth}=2$. Ridge's L2 penalty shrinks coefficients toward zero uniformly, acting as an implicit Bayesian prior. The minimal 3-feature set eliminates the curse of dimensionality entirely—each feature has 39 observations per parameter.

Cross-Variable Coherence with Other Forecasts

The NFP forecast is coherent with the unemployment and wage signals from the other qualified models:

- **NFP above consensus + Unemployment below consensus** ($\rightarrow$ tighter labour market): The NFP model sees 127K vs 95K consensus; the unemployment model forecasts 4.05% vs 4.3% consensus ($P(\text{miss})=60\%$). Both independently point to a labour market that is stronger than the market expects.
- **Stronger hiring + Higher wage growth** ($\rightarrow$ Phillips curve channel): The AHE model forecasts 0.41pp MoM vs 0.2pp consensus ($P(\text{beat})=67\%$). A tighter labour market (from NFP/unemployment signals) mechanically supports stronger wage bargaining—exactly what the AHE model detects.
- **Implication for rates:** The three US labour signals are unanimously hawkish. If realised, this combination would: (i) push Fed rate cut expectations further out, (ii) steepen the front end of the Treasury curve (higher terminal rate), and (iii) spill into global duration via the VAR channels identified in our spillover analysis—particularly UK gilts (29% of Level variance from US) and Canadian bonds (47% of Slope variance from US).
- **Brazil IPCA miss as a partial offset:** The BR model ($P(\text{miss})=79\%$) provides a disinflationary counterpoint in EM. A below-consensus IPCA print would ease pressure on the BCB and compress DI rates, partially offsetting the hawkish US impulse for global risk sentiment.

Next-Period Forecasts: May 8, 2026 Releases

Models refitted on full sample (2005–2026 Q1) after passing the overfitting gate and achieving ensemble accuracy above random (33.3%). Only qualified models are used for inference.

Point Forecasts & Confidence Intervals

Variable	Consensus	XGB Point	QR Median	50% CI	80% CI
US Unemployment (%)	4.30	4.05	4.17	[3.74, 4.30]	[3.70, 4.53]
US AHE MoM (pp)	0.20	0.41	0.73	[0.59, 1.52]	[0.48, 1.58]
BR IPCA YoY (%)	4.70	0.61	0.54	[0.32, 0.70]	[0.30, 0.75]

Beat/Miss Probabilities (Ensemble: 25% OL + 50% XGB + 25% QR)

Variable	P(miss)	P(inline)	P(beat)
US Unemployment	59.8%	27.9%	12.3%
US AHE MoM	1.2%	31.8%	67.0%
BR IPCA YoY	79.4%	9.6%	11.0%

Interpretation

- **US Unemployment (P(miss)=60%):** The model forecasts 4.05–4.17% vs. 4.3% consensus. This implies a tighter labour market than expected — consistent with the still-elevated yield curve slope (restrictive policy not yet biting through to employment). The 80% CI spans 3.70–4.53%, so the miss signal is moderate rather than decisive.
- **US AHE MoM (P(beat)=67%):** Strong upside surprise expected. The XGB point (0.41pp) and QR median (0.73pp) both exceed the 0.2pp consensus. This is coherent with the unemployment miss — a tighter labour market drives stronger wage growth (Phillips curve). *This would be hawkish for the Fed and bearish for Treasuries.*
- **BR IPCA YoY (P(miss)=79%):** The ensemble strongly favours a downside surprise. BRL has stabilised and DI curve levels have compressed, both of which the model interprets as disinflationary signals. A miss would support the case for BCB easing later in the cycle and compress the long end of the DI curve.

Cross-Variable Coherence

The US unemployment miss and AHE beat are internally consistent: both point to a labour market that is tighter than consensus expects. Together, they suggest a hawkish data print on May 8, which would be positive for USD, negative for US duration, and potentially spill into higher gilt yields (per our VAR spillover analysis).

Data: FRED (23 US/intl macro series), IPEA (Brazil). **Models:** XGBoost, Ordered Logit (MLE), Quantile Regression. **Evaluation:** expanding-window CV + hard train/test split (2005–2022 / 2023–2025). Forecasts generated from models refitted on full available sample after passing test-set validation. This document does not constitute investment advice.